BTS-VI(SS)-01.16-0053 Reg.No. | | | | | | | | |

**B**

# *B.Tech. Degree VI Semester Special Supplementary Examination January 2016*

**ME 1603 OPERATIONS MANAGEMENT**

*(2012 scheme)*

Time : 3 Hours Maximum Marks : 100

## PART A

(Answer *ALL* questions)

$(8 \times 5 = 40)$

I. (a) What are the three time estimates of an activity in PERT? How is the expected time of an activity calculated?

(b) What are the different types of forecasting methods?

(c) Describe "Infinite delivery rate without back ordering"

(d) Define "dispatching" and "expediting".

(e) What is Gantt chart? Name any two.

(f) What is the basic function of Aggregate planning process?

(g) Briefly discuss any two space determination methods.

(h) List out the reasons for replacement of equipment.

## PART B

$(4 \times 15 = 60)$

II. A project consists of the activities as given in the table below.

| Job | Duration(weeks) | | |
|---|---|---|---|
| | Optimistic | Most likely | Pessimistic |
| 1 - 2 | 3 | 6 | 15 |
| 1 - 6 | 2 | 5 | 14 |
| 2 - 3 | 6 | 12 | 30 |
| 2 - 4 | 2 | 5 | 8 |
| 3 - 5 | 5 | 11 | 17 |
| 4 - 5 | 3 | 6 | 15 |
| 6 - 7 | 3 | 9 | 27 |
| 5 - 8 | 1 | 4 | 7 |
| 7 - 8 | 4 | 19 | 28 |

(i) Draw the project network.

(ii) Find the expected length and variance of the project.

(iii) What is the probability that the project will be completed in 41 weeks?

**OR**



**(P.T.O.)**

III.

| Month | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Sales (units) | 180 | 168 | 159 | 170 | 188 |

Estimate the sales forecast during the 5 months for the data given in the above table, taking $\alpha = 0.5$ and the forecast for the first month as 160 units.

IV. Enumerate the objectives and functions of PPC, with suitable examples.

**OR**

V. Explain "stores ledger" and "BIN cards" and their uses with the help of diagrams.

VI. Consider the following two machines and 5 job problems. Obtain the optimal schedule and the corresponding makes span.

| Job | Process time in hrs. | |
|---|---|---|
| | Machine – 1 | Machine - 2 |
| 1 | 5 | 7 |
| 2 | 10 | 13 |
| 3 | 8 | 11 |
| 4 | 9 | 12 |
| 5 | 6 | 4 |

**OR**

VII. (a) Explain the following rules in scheduling with simple examples of your choice.
   (i) First come first serve.
   (ii) Shortest processing time.
   (iii) Earliest due date.

(b) What is critical ratio method in sequencing?

VIII. Which are the various types of plant layout? Describe all in detail.

**OR**

IX. List out the principles of material handling and explain the selection factors of material handling equipments.

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